Minimum Distance from Point to Line

11. For each of the following, find the distance from $Q(3, -2)$ to each line.

a. $3x - 2y - 6 = 0$

b. $\frac{x-3}{2} = \frac{y-4}{7}$

c. $\vec{r} = (-3, -7) + t\left(\frac{1}{5}, \frac{1}{6}\right)$

d. $x = -5$

12. Find the distance from each of the following points to the line $6x + 3y - 10 = 0$.

a. $(4, 7)$ b. $(4, -8)$ c. $(0, 5)$ d. $\left(5, -\frac{20}{3}\right)$

15. a. Prove that the distance from a point Q in space to a line through a point P with direction vector $\vec{d}$ is equal to $\frac{|\overrightarrow{PQ} \times \vec{d}|}{|\vec{d}|}$.

b. Find the distance from the point $Q(1, -2, -3)$ to the line $\vec{r} = (3, 1, 0) + t(1, 1, 2)$.

c. Find the distance between the parallel lines $\vec{r} = (-2, 2, 1) + t(7, 3, -4)$ and $r = (2, -1, -2) + u(7, 3, -4)$.

Answers

11.a) $\frac{7\sqrt{13}}{13}$ b) $\frac{12\sqrt{53}}{53}$ c) 0 d) 8

12.a) $\frac{7\sqrt{5}}{3}$ b) $\frac{2\sqrt{5}}{3}$ c) $\frac{\sqrt{5}}{3}$ d) 0

15.b) $\frac{\sqrt{66}}{6}$ c) $\frac{\sqrt{115070}}{74}$